Week 1 Presentation Content

Project framing, dataset approval and delivery plan

Presentation goal

Secure approval for the problem, dataset, decision questions, analytical methods, architecture and four-week delivery plan.

Deck map

Slide	Purpose
1	Weekly status and time plan
2	Project-specific SWOT
3	Team communication plan
4	Problem, stakeholders and objective
5	Dataset and decision questions
6	Method selection
7	System architecture
8	Four-week work breakdown
9	Research paper plan and approval request

Evidence to keep open during Q and A

- requirement/Week_1.pdf and requirement/Big Data Analytics_Final_Project_2026_VNUK.pdf
- MovieLens_10M_Final_Project_Plan.md and README.md
- Dataset README with 10,000,054 ratings, 10,681 movies and 71,567 total users

How to use this file Copy the on-slide text into PowerPoint. Use the visual direction to build the graphic, and rehearse from the speaker cue. Update blank actual-effort fields before presenting.

Slide 1 This week establishes the project contract

On-slide copy

- WHAT: define a decision problem, approve the MovieLens 10M dataset, choose methods and assign four weeks of work.
- WHEN: complete scope, architecture and paper outline before data engineering begins in Week 2.
- WHY: later modelling and dashboard choices need one agreed decision frame and reproducible pipeline.
- PLANNED EFFORT: 8-12 team hours. ACTUAL EFFORT: _____ hours. GAP: _____ hours.
- ACTION: record any scope change, owner and due date before closing the meeting.

Visual direction Use a horizontal What / When / Why / Time strip. Keep actual effort blank until the team confirms it.

Speaker cue Open with the decision the class should approve, then explain why the four-week sequence is feasible.

Slide 2 SWOT is specific to this project

On-slide copy

- Strengths: 10 million ratings, a clear relational model, reproducible SQL and KNIME workflows, and model outputs written back to MySQL.
- Weaknesses: historical data ends in 2009; there is no exposure, viewing, revenue or demographic data.
- Opportunities: segment-based curation, discovery campaigns, title-level promotion tests and model governance.
- Threats: target leakage, slow full-table operations, cold-start exclusions and overclaiming business impact from ratings alone.

Visual direction Build a 2-by-2 SWOT matrix with one short line per item.

Speaker cue Connect each weakness or threat to a control later in the deck: chronological splits, TRAIN-only features, support thresholds and stated limitations.

Slide 3 Communication plan keeps work auditable

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- Project owner: scope, weekly status, decisions and final integration.
- Data and SQL owner: schema, import, validation queries and dashboard views.
- KNIME and modelling owner: preprocessing, K Means, Logistic Regression and writeback.
- Dashboard and report owner: Power BI pages, evidence chain, IEEE paper and presentation assembly.
- Cadence: one midweek checkpoint, one end-of-week demo and GitHub updates after each approved change.

Role	Member assignment	Primary handoff
Project owner	_____	Approved scope and weekly status
Data and SQL	_____	Validated tables and SQL evidence
KNIME and ML	_____	Workflows, metrics and writeback
Dashboard and paper	_____	Decision views and final narrative

Visual direction Use a RACI-style lane diagram. Add team names only after roles are agreed.

Speaker cue State the escalation rule: data or model changes that affect the decision questions must be raised at the next checkpoint.

Slide 4 The project supports content promotion decisions

On-slide copy

- Problem: broad popularity lists can hide titles that fit distinct audience segments.
- Primary stakeholders: content strategists, campaign planners, analysts and model reviewers.
- Objective: combine audience segments and positive-engagement probabilities to identify titles to promote, maintain, reduce or test.
- Success means a reproducible path from raw MovieLens files to evidence-backed dashboard decisions.

Visual direction Show a single decision chain: audience signal -> model evidence -> dashboard -> action.

Speaker cue Avoid claiming revenue lift. The dataset supports preference and engagement analysis, not causal business impact.

Slide 5 MovieLens 10M is large enough for the analytical stack

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- 10,000,054 ratings from 69,878 rating users across 10,681 movies.
- 95,580 tags and 21,564 movie-genre relationships enrich content context.
- Ratings use a 0.5-5.0 scale; the observed average is 3.512 and 50.06% of ratings are at least 4.0.
- The time range is 9 January 1995 to 5 January 2009, which supports chronological validation.
- Decision questions: Which audience segments exist? Which users are likely to rate positively? Which titles have high predicted appeal but lower historical popularity? Which results are reliable enough to act on?

Visual direction Use four KPI tiles plus the four decision questions beneath them.

Speaker cue Justify the dataset with scale, relational structure and timestamps; immediately note its historical and non-commercial limits.

Evidence Dataset README and SQL quality-check output.

Slide 6 Two models answer different decision questions

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- K Means groups warm-start users by activity, rating behavior and genre affinity.
- Logistic Regression estimates whether a rating will be positive, defined as rating >= 4.0.
- K Means supports audience strategy; Logistic Regression supports title-level prioritization and transparent evaluation.
- A chronological TRAIN / VALID / TEST design limits future information from entering historical features.

Visual direction Use a two-column method-to-decision map. Put K Means on the left and Logistic Regression on the right.

Speaker cue Explain that method choice follows the decision question: segmentation for audience structure, classification for positive engagement.

Slide 7 The architecture separates computation from decisions

On-slide copy

- MovieLens files -> MySQL raw and curated tables -> KNIME preprocessing and modelling.
- KNIME writes segments, predictions and model metrics back to MySQL.
- Power BI reads versioned result tables and views for five decision-oriented pages.
- GitHub stores SQL, workflow documentation, dashboard assets and the research paper.

Visual direction Use tmp/report/figures/architecture.png as the base visual.

Speaker cue Walk left to right. Emphasize that MySQL is both the governed input layer and the model-output contract.

Evidence Repository architecture plan, SQL scripts, KNIME assets and dashboard PBIP files.

Slide 8 The work breakdown follows four evidence gates

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- Week 1: approve scope, dataset, decision questions, architecture and paper outline.
- Week 2: validate data, build the MySQL schema, connect KNIME and create the modelling dataset.
- Week 3: engineer leakage-safe features, evaluate both models, write results to MySQL and draft the dashboard.
- Week 4: finish the dashboard, recommendations, reproducible repository, IEEE paper and Q and A.

Week	Planning baseline	Exit evidence
1	8-12 team hours	Approved project contract
2	16-20 team hours	Validated SQL-KNIME pipeline
3	20-24 team hours	Evaluated ML and dashboard draft
4	16-20 team hours	Final system, paper and presentation

Visual direction Use a four-lane timeline with an evidence gate at the end of each week.

Speaker cue Treat these hours as a planning baseline, not reported effort.

Slide 9 The paper documents one reproducible decision system

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- Working title: Segment-Based Movie Promotion and Content Discovery Strategy Using MovieLens 10M.
- Outline: Abstract, Introduction, Related Work, Methodology, Experiments and Results, Discussion and Recommendations, Conclusion, References.
- Initial references: MovieLens dataset documentation; Harper and Konstan on MovieLens; Lloyd on least-squares clustering; official MySQL, KNIME and Power BI documentation.
- Approval requested: problem, dataset, four decision questions, K Means plus Logistic Regression, architecture and four-week plan.

Visual direction End with a checklist that the instructor can approve verbally.

Speaker cue Ask for specific objections to scope or method. Record changes as decisions, owners and dates.

Evidence Final-project brief requires an IEEE-style paper of 6-10 pages with at least five sources.